1.09 Quadratics: Problem Solving

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2: Completing the Square

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"Completing the square" means putting a quadratic in the form $(x+a)^2+b$ or $a(x+b)^2+c$

The **underlying reason we do this is because x only appears once in the expression** (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

If we have a completed square:

$$(x+4)^2-7=0$$

we saw at the start of the chapter how we could rearrange to make x the subject.
Indeed using the quadratic formula is actually solving the quadratic by completing the square – it's just someone has done the work for us already!

b. Sketching Quadratics

We'll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to 'integrate' expressions like:

$$\int \frac{1}{x^2-4x+5}\,dx$$

(you will cover integration later this module)

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Completing the Square Recap

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Expand:

$(x+\mathbf{9})^2 = x^2+$

$(x-\mathbf{5})^2 = x^2-$

What do you notice about the relationship between the bold numbers?

1.09 Quadratics: Problem Solving

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Completing the Square

13

Express $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

$= 2\left(x\right.$

$= 2\left(\left(\right.\right.$

$= 2(x$

$= 2(x$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x + c)^2$

$= -3x^2$

$= -3\left(x\right.$

$= -3\left(\left(\right.\right.$

$= -3(x$

$= 8 - 3($

It may help to write in the form $ax^2 + bx + c$ first.

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Test Your Understanding

14

Express $3x^2 - 18x + 4$ in the form $a(x + b)^2 + c$

$= 3\left(x\right.$

$= 3\left(\right.$

$= 3(x$

$= 3(x$

Express $20x - 5x^2 + 3$ in the form $a - b(x + c)^2$

$= -5$

$= -5$

$= -5$

$= -5$

$= 23$

1.09 Quadratics: Problem Solving

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Solving by Completing the Square

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Solve the equation:

$$3x^2 - 18x + 4 = 0$$

x^2

$(x$

$(x$

x

x

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Exercise 2C/2D Pure Y1 Page 23-24

16

1 Complete the square for the expressions: **a** $x^2 + 4x$ **b** $x^2 - 6x$ **c** $x^2 - 16x$ **d** $x^2 + x$ **e** $x^2 - 14$

2 Complete the square for the expressions: **a** $2x^2 + 16x$ **b** $3x^2 - 24x$ **c** $5x^2 + 20x$ **d** $2x^2 - 5x$ **e** $8x - 2x^2$

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4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\mathrm{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \mathrm{and} B$ are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

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Proving the Quadratic Formula

18

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$ax^2 + bx + c = 0$

$x^2 + \frac{b}{a}$

$\left(x + \frac{b}{2}\right.$

$\left(x + \frac{b}{2}\right.$

$\left(x + \frac{b}{2}\right.$

$x + \frac{b}{2a}$

$x = \frac{-b}{}$

Just use exactly the same method as you usually would!

1.09 Quadratics: Problem Solving

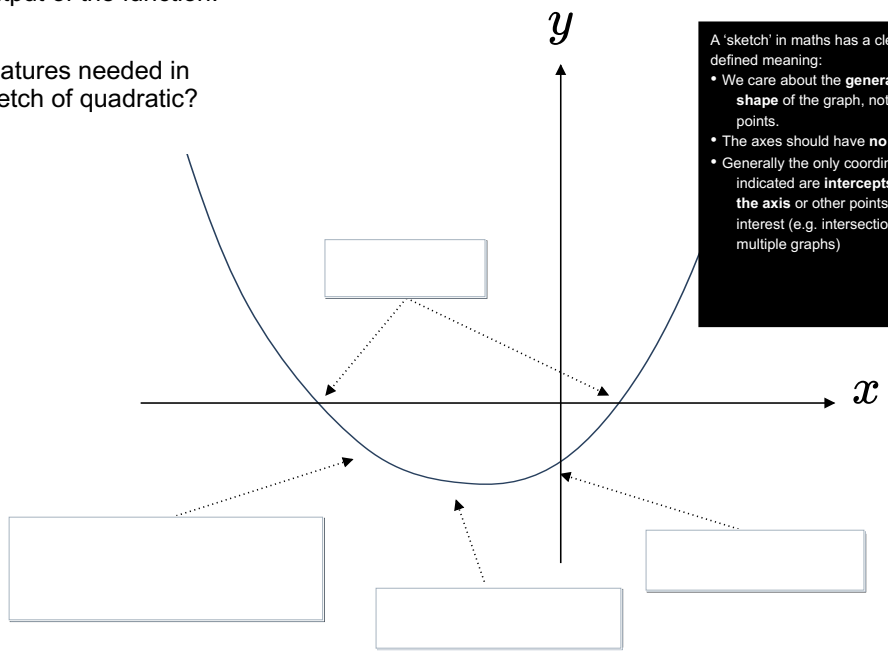
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4: Quadratic Graphs

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Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

Features needed in sketch of quadratic?



A 'sketch' in maths has a clearly defined meaning:

- We care about the **general shape** of the graph, not exact points.
- The axes should have **no scale**.
- Generally the only coordinates indicated are **intercepts with the axis** or other points of interest (e.g. intersections of multiple graphs)

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Example

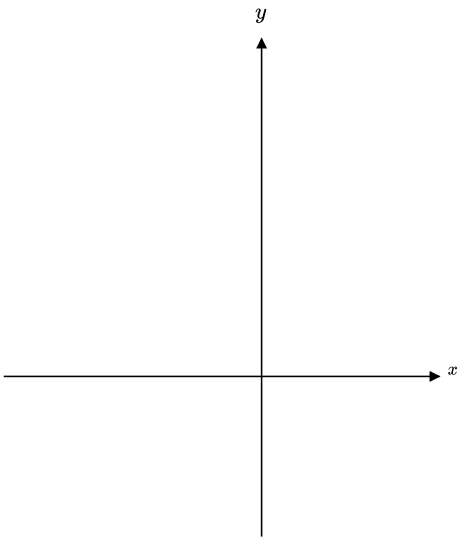
27

Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

y-intercept:

Turning point:



Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

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Example

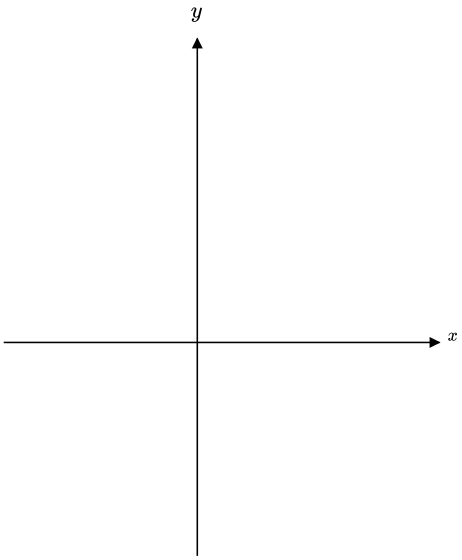
28

Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

Roots:

y-intercept:

Turning point:



Line of reflection:

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Test Your Understanding

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Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a $y = x^2 + 4$

b $y = x^2 - 7x + 10$

c $y = 5x + 3 - 2x^2$

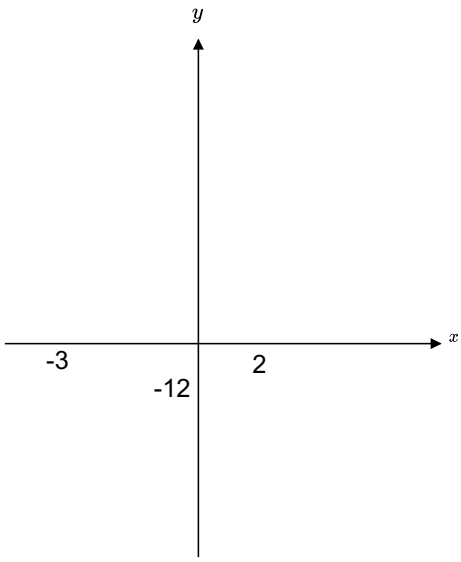
d $y = x^2 + 4x + 11$

1.09 Quadratics: Problem Solving

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Determining the Equation using a Graph

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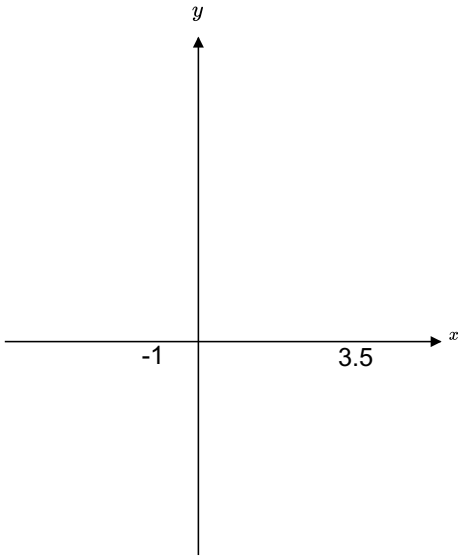


Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

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Further Example

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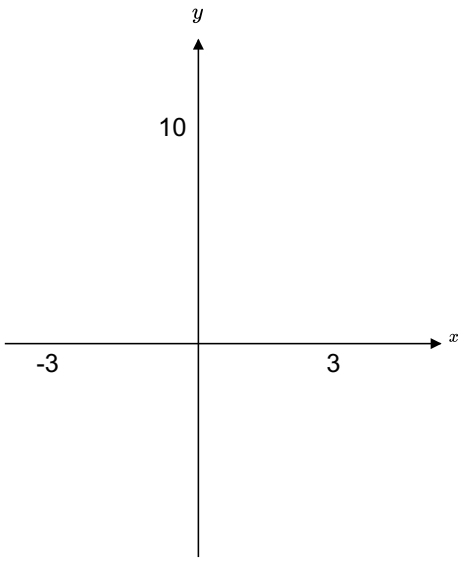
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

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Test Your Understanding

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Determine an equation of this quadratic graph.

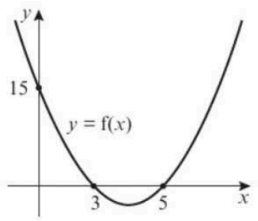
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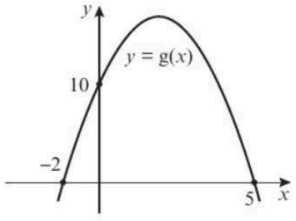
33

2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a, b, c for each function.

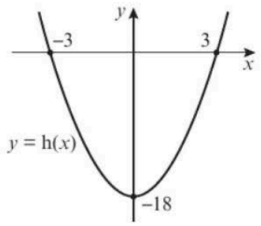
a



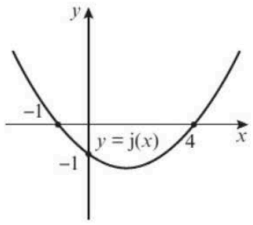
b



c



d



1.09 Quadratics: Problem Solving

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Exercise 2F Pure Y1 Page 30

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3 The graph of $y = ax^2 + bx + c$ has a minimum at $(5, -3)$ and passes through $(4, 0)$.
Find the values of a, b, c.

Extension Question:
[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?
$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

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Starter

35

How many **distinct** real solutions do each of the following have?

$x^2 - 12x + 36 = 0$
$x^2 + x + 3 = 0$
$x^2 - 2x - 1 = 0$

1.09 Quadratics: Problem Solving

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5: The Discriminant

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$$ax^2 + bx + c = 0$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

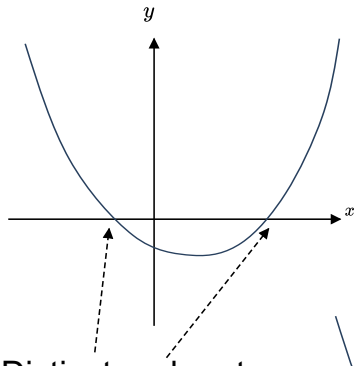
Note: Roots of a **function** f are the values of x such that $f(x) = 0$.
Similarly the roots of an **equation** are solutions to an equation in the form $f(x) = 0$

Looking at this formula, when in general do you think we have:
- No real roots?
- Equal roots?
- Two distinct roots?

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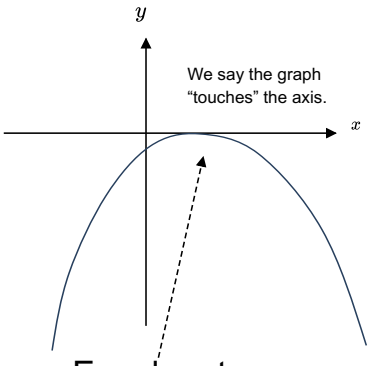
5: The Discriminant

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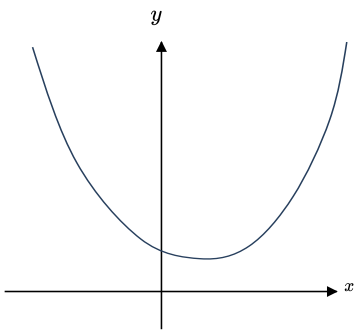
Distinct real roots

$\mathbf{b^2 - 4ac > 0}$



Equal roots

$\mathbf{b^2 - 4ac = 0}$



No real roots

$\mathbf{b^2 - 4ac < 0}$

1.09 Quadratics: Problem Solving

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Just for your interest…

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Why do we say “Equal Roots” not “One root”?

$$x^2 - 12x + 36 = 0$$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say “*it has one repeated root*” or “*it has equal roots*”, i.e. indicating we have **2 roots** (but with the same value). Why not say it has 1 root?

It is due to the **Fundamental Theorem of Algebra**:

“Every polynomial of order n has exactly n roots.”

A **polynomial** is an expression with non-negative integer powers of x, i.e. $a + bx + cx^2 + dx^3 + \ldots$ All linear, quadratic and cubic expressions are examples of polynomials.

The **order** of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

These roots might be repeated or might not be ‘real’ roots. $\sqrt{-1}$ is known as a **complex number**, which you will encounter if you do FM. But **it is still a value**!

The theorem means that a quadratic (order 2) will **always** have 2 roots. This is why you should say “no **real** roots” when $b^2 - 4ac < 0$ rather than “no roots”, because there are still roots – it’s just they’re not ‘real’! Similarly we must say “*equal roots*” because there are still 2 roots.

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by **Argand**. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is **provably** no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

There are various other ‘Fundamental Laws’. The ‘**FL of Arithmetic**’ you encountered at KS3, which states that “*every positive integer* > 1 *can be written as a product of primes in one way only*”. You will encounter the ‘**FL of Calculus**’ in Chapter 13.

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Quickfire Questions

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Equation	Discriminant	Number of Distinct Real Roots
$x^2 + 3x + 4 = 0$		
$x^2 - 4x + 1 = 0$		
$x^2 - 4x + 4 = 0$		
$2x^2 - 6x - 3 = 0$		
$x - 4 - 3x^2 = 0$		
$1 - x^2 = 0$		

1.09 Quadratics: Problem Solving

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Problems involving the discriminant

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8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a) $a =$ $b =$ $c =$ 4

$(2p)^2 - 4$

$4p^2 - 12p$

$p^2 - 3p -$

$(p + 1)(p$

$p = 4$

b)

$(x$

Tip: Always start by writing out a, b and c explicitly.

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Exercise 2G Pure Y1 Page 32

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2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

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4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

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6 The function $\mathrm{g}(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p.

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

7 $\mathrm{h}(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $\mathrm{h}(x)$ in terms of k.

b Hence or otherwise, prove that $\mathrm{h}(x)$ has two distinct real roots for all values of k.

b

1.09 Quadratics: Problem Solving

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Test Your Understanding

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$$x^2 + 5kx + (10k + 5) = 0$$

where k is a constant.
Given that this equation has equal roots, determine the value of k.

$a(25(k$

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

$a33k$

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Exercise 2G Pure Y1 Page 32

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Extension Questions:

1 [MAT 2009 1C] Given a real constant c, the equation

$$x^4 = (x - c)^2$$

Has four real solutions (including possible repeated roots) for:
A) $c \leq \frac{1}{4}$
B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$
C) $c \leq -\frac{1}{4}$
D) all values of c

Square rooting: $\mathbf{x^2 = \pm(x - c)}$

2 [MAT 2006 1B] The equation $\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?
A)

3 [MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:
A) $P^3 > A$
B) $A^2 > 2P + 1$
C) $P^2 \geq 16A$
D) $PA > A + P$
E)

Let x and y be the width and height.
A = xy, P = 2x + 2y